

CYBERSECURITY FUNDAMENTALS

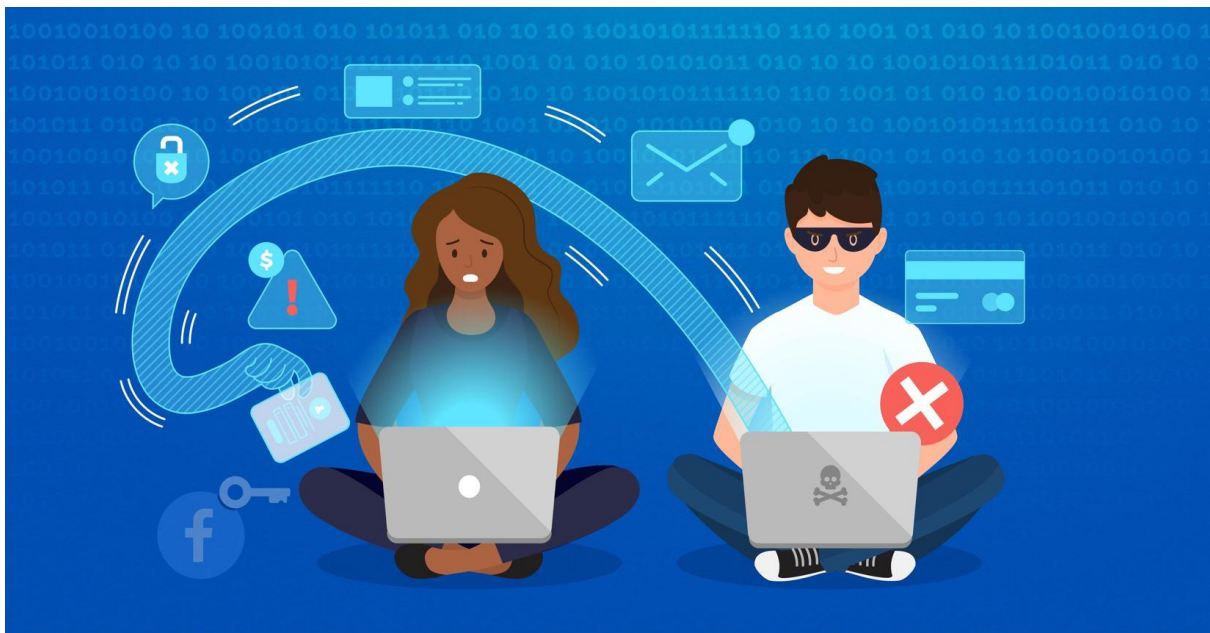
Introduction

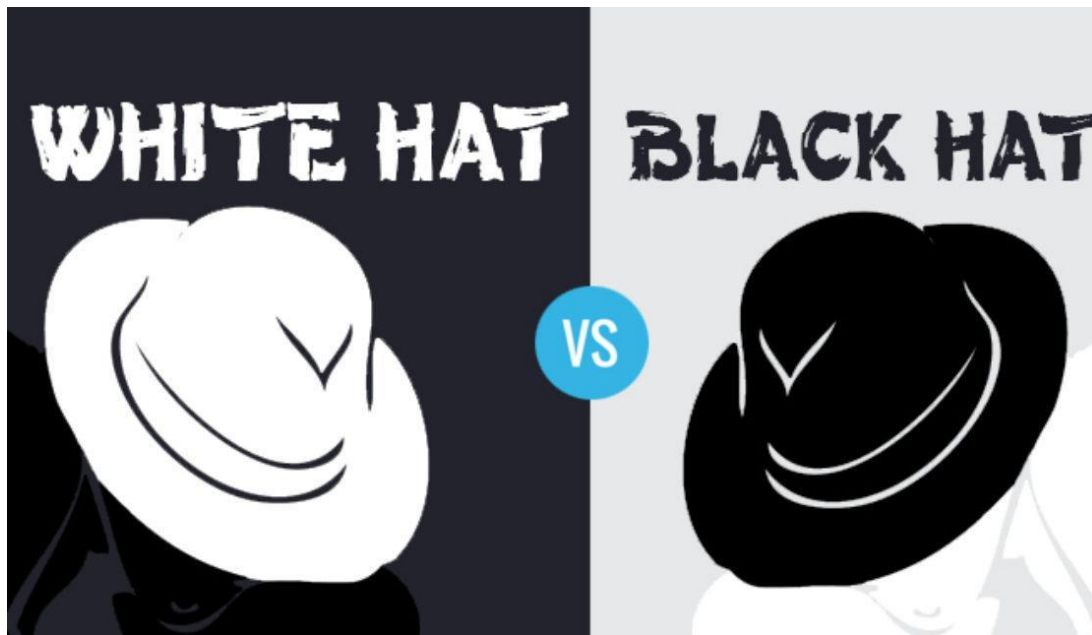
Cybersecurity is the practice of protecting computer systems, networks, and data from digital attacks, unauthorized access, and damage. It ensures the safety and privacy of information in the digital world.

Ethical Hacking

Ethical hacking is the process of testing systems for security vulnerabilities in a legal way. Ethical hackers help organizations identify and fix weaknesses before attackers can exploit them.

Types of Hackers



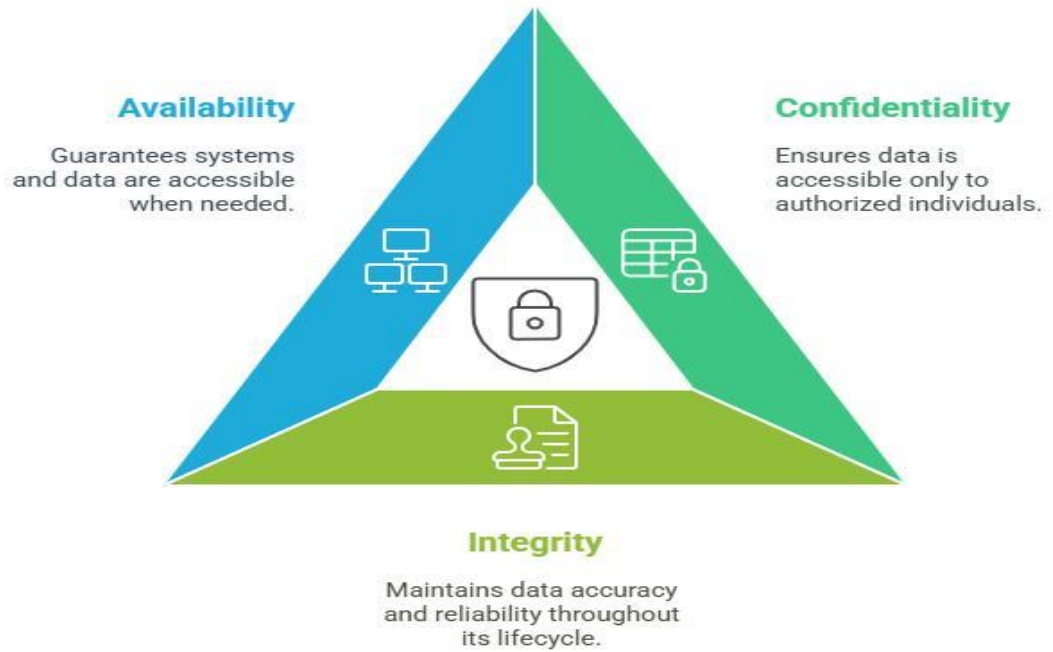


- **White Hat Hackers** – Ethical and legal hackers
- **Black Hat Hackers** – Malicious hackers
- **Grey Hat Hackers** – Between ethical and unethical

CIA Triad



The CIA Triad



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The CIA Triad is a core concept in cybersecurity:

- **Confidentiality** – Protecting data from unauthorized access
- **Integrity** – Ensuring data is accurate and not altered
- **Availability** – Ensuring data is accessible when needed

Common Cyber Threats

5 Common Cybersecurity Threats

and Why They're Dangerous



Phishing

Fake emails or websites that trick users into revealing sensitive information

⚠ Most common entry point for attacks that can fool even experienced users



Malware

Malicious software that damages systems or steals data

⚠ Corrupts files, steals information, and grants attackers control



Ransomware

Malware that encrypts files and demands payment to restore access

⚠ Stops business operations; paying doesn't guarantee data recovery



Internet-Facing Vulnerabilities

Exposed system weaknesses that attackers exploit remotely

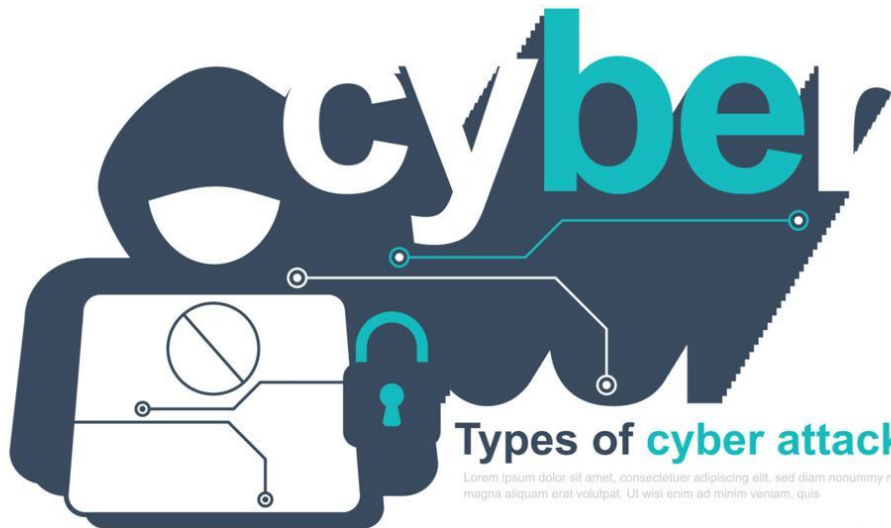
⚠ Allows attackers to gain network access without user action



Business Email Account Takeover

When attackers hijack a legitimate email account to impersonate employees

⚠ Causes major financial and data losses



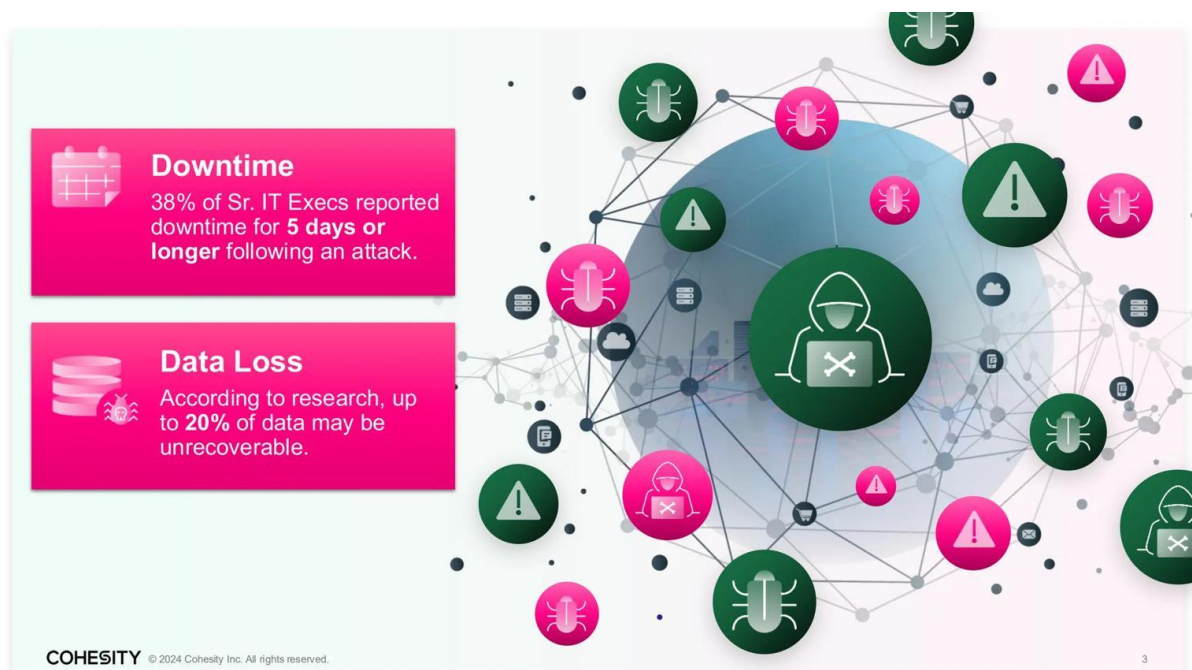
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Malware VS Ransomware

	Malware	Ransomware	
TYPES	 2 Main Type: Lock / Encrypt	 Multiple Types	Ransomware attacks rose by 151% in 2021
VECTOR	 Phishing is the #1 Delivery Vehicle	 USB Devices Risky Websites Phishing Software Downloads	“Ransomware is one of the greatest concerns when it comes to cyber threats” -World Economic Forum
RECOVERY	 Restore encrypted data from a secure backup or Pay the ransom	 Antivirus, security expert, reformat/reinstall, recover from a secure backup	%97 of ransomware enters companies through phishing emails
POTENCY	 In 2021 a ransomware attack took place every 11 seconds. Damage of \$20 billion	 Antivirus, security expert, reformat/reinstall, recover from a secure backup	“Ransomware is the Greatest Business Threat in 2022” -Nasdaq

- **Malware** – Harmful software that damages systems
- **Phishing** – Fake emails/websites to steal information
- **Ransomware** – Locks files and demands money

Impact of Cyber Threats





Cyber attacks can cause:

- Data loss
 - Financial loss
 - Privacy breaches
 - System downtime
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Conclusion

Cybersecurity is essential for protecting digital systems and information. Understanding threats and security principles helps prevent cyber attacks and ensures a safe digital environment.

Author

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